MIL-P-50526A (PA)
23 May 1974
SUPERSEDING
MIL-P-50526 (MU)
15 April 1970

MILITARY SPECIFICATION

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PROJECTILE, 8 INCH, HE: M404,

BODY ASSEMBLY AND BASE PLUG FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for the body assembly and base plug for Projectile, 8 Inch, HE: M404.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition, General Specification for
MIL-S-20426	-	Shells, Base Ejection and Illuminating Type (Other Than Mortar Shell), Metal Parts For
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC: 1320

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection By Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables For Inspection By Attributes

DRAWINGS

U. S. ARMY ARMAMENT COMMAND

F-7548346	-	Pallet for 8" Projectile, Empty or Loaded
D-8875892	-	Plug, Base
B-8875893	-	Gasket,Base Plug
F-8875940	-	Body Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 8-69	-	Tension Testing of Metallic Materials
ASTM B 117-73	-	Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Body Assembly and Base Plug.-The body assembly and base plug shall comply with all requirements on Drawings (Dwgs) F-8875940 and D-8875892, respectively, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Rotating band clearance (tightness).-The clearance between the band and the band seat in a radial direction shall not exceed 0.006 inch.

3.4 Tension test of rotating band.-The rotating band, after removal from the body assembly, shall comply with the tensile strength and percentage of elongation requirements specified in MIL-S-20426 when tested as specified in 4.4.4.

3.5 Bend test of rotating band.-The rotating band, after removal from the body assembly, shall comply with bend test requirements specified in MIL-S-20426 when tested as specified in 4.4.5.

3.6 Proving ground ballistic tests.-The rounds shall function satisfactorily in accordance with the intent of the design and shall show no evidence of metal parts failure when tested as specified in 4.4.9. The following conditions shall be considered as evidence of unsatisfactory functioning or metal parts failure:

(a) Loss, separation, breakup, or fracture of metal parts in gun bore or during flight.
(b) Shear or loss of rotating band.
(c) Metal parts failure during cargo-expulsion.
(d) Evidence of movement of boattail on the forward body.
(e) Failure of round to reach impact area. (short round).
(f) Failure of round to eject cargo.

3.7 Use of steel blooms or billets with hot-sheared ends (applicable when this process is used).-The contractor shall demonstrate his ability to produce acceptable body assemblies and base plugs manufactured from semi-finished forging quality, carbon steel blooms or billets with hot-sheared ends. This shall be demonstrated by satisfactorily producing the first article quantities (as evidenced by successful completion of

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all first article examinations and tests) and satisfactorily producing six thousand (6000) body assemblies and base plugs as first deliveries under the contract from the above-stated material.

3.8 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Discoloration, not due to corrosive action, will be permitted. All required marking shall be neat and sharply defined.

3.8.1 Heat treatment.-All heat treatment procedures shall be controlled so as to produce in each lot of components the mechanical properties specified on the applicable component drawings and in applicable specifications. (See 4.2.4)

3.9 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.10 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review. The certification covering the groove pins (Dwg. C-8875885) shall include a statement that they comply with all requirements for hardness, salt spray, and double shear strength for each heat treatment lot of pins.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Ten (10) base plug gaskets.
b. Ten (10) unpainted base plugs with protective coating.
c. Five (5) painted base plugs.
d. Ten (10) rotating band blanks.
e. Ten (10) boattails (unpainted).
f. Ten (10) forward bodies.

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g. (Five (5) forward body and aleeve assemblies (Dwg. F-8875937).

h. Ten (10) unbanded, unpainted body assemblies, with protective coating.

i. Ten (10) unpainted body assemblies, with protective coating.

j. Ten (10) painted body assemblies.

k. Thirty-six (36) completed, painted body assemblies and thirty-six (36) painted base plugs, with thirty-six (36) gaskets. (for ballistics testing)

All parts and completed assemblies submitted shall have been produced by the contractor, or furnished by a supplier, using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance for materials. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings. In addition, the following tests shall be performed:

4.2.2.1 Proving ground ballistic tests (See 3.6).-The thirty-six (36) painted body assemblies specified in 4.2.1 (k) shall be forwarded, with base plugs, to the loading facility designated by the Contracting Officer.

The loading facility shall load, assemble and pack the body assemblies as specified in 6.8. The projectiles shall be forwarded to the Government-owned proving ground specified by the Contracting Officer and be subjected to the following ballistics tests using the methods and procedures specified in 4.4.9.

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4.2.2.1.1 Ten (10) of the loaded assemblies shall be fired at Zone 1 (See 4.4.9.1).

4.2.2.1.2 Ten (10) of the loaded assemblies shall be fired at Zone 7. Five (5) of these assemblies shall first be subjected to a five-foot drop test. (See 4.4.9.2)

4.2.2.1.3 Fifteen (15) of the loaded assemblies shall be fired at Zone 7X. (See 4.4.9.3)

4.2.3 Rejection.-If any body assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted (including successful completion of all ballistic tests), the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Mechanical properties.-Documentation detailing all equipment, operations, and procedures to be used during the term of this contract, and affecting one or more of the mechanical properties shall be submitted as part of first article inspection. This documentation, with actual test data, shall be submitted to the technical agency specified in 6.6. Changes in the established procedures or equipment during the course of the contract shall be documented and submitted, with test data, as specified in 6.6. Documentation shall include actual chemical and metallurgical analyses of material, including wrought materials (hardness, grain size and grain structure with photographs). The finished items shall also provide data for tensile strength, yield strength, percent elongation, percent reduction in area and impact strength. Documentation covering normalizing, tempering, stress-relieving and annealing, when any of these operations are used, shall provide the shortest and longest time at the temperature for each item, size and type of furnace or oven, and the load and distribution of the items in the furnace or oven. When quenching is employed, documentation shall indicate the quenching medium, its temperature, method of presentation of items into the medium and length of time in the medium. This documentation shall be signed by a responsible agent of the prime contractor.

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4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.1.1 Heat treatment or stress-relief lot.-A heat treatment or stress-relief lot shall consist of bodies and boattails from one mill-heat of steel that have been heat-treated or stress-relieved in the same furnace at the same temperature and for the same length of time. Test samples shall be selected after completion of all heat treatment operations. Where heat-treatment or stress-relief is done in a continuously operated, conveyor-type furnace, the identification maintained shall be that code number assigned the particular mill-heat of steel. Where heat treatment or stress-relief is performed in batch-type furnaces, additional identification control must be maintained that directly relates to that portion of the mill-heat processed as a batch. The method of control must meet with the approval of the Government. Components in heat treatment or stress-relief lots shall remain grouped together for identifying purposes until mechanical properties tests have been completed. Components from heat-treatment or stress-relief lots which comply with the applicable requirements may be subsequently regrouped into body assembly lots. Heat treatment lots shall be suitably identified by an approved letter or number system. Coding by color shall not be used.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Gasket, base plug (See Dwg. B-8875893).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major: (See 4.3.3.1)			
101.	Inside diameter	Gage	01001
102.	Outside diameter	Gage	01002
Minor:			
201.	Thickness	Gage	01003

4.3.2.2 Sleeve (See Dwg. C-8875890 covering a detail of Dwg. F-8875937).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Interior finish improper	Visual	02001
Minor:			
201.	Knurl missing	Visual	02002
202.	Chamfer missing or improper (when applicable)	Visual	02003
203.	Evidence of poor workmanship (See 3.8)	Visual	02004

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4.3.2.3 Body, forward (prior to protective coating) (See Dwg. F-8875889 covering a detail of Dwg. 8875937).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Chamfer at rear end, min (See 4.3.3.2)	Gage	03001
102.	Pitch diameter of rear thread	Gage	03002
103.	Minor diameter of rear thread, max.	Gage	03003
104.	Concentricity of largest (sleeve)cavity diameter with pitch diameter of rear thread (including bottom of cavity before sleeve seat shoulder)	Gage	03004
105.	Distance from nose end to base of larger cavity (when applicable)	Gage	03005
106.	Depth of large cavity	Gage	03006
107.	Diameter of smaller cavity	Gage	03007
108.	Diameter of large cavity	Gage	03008
109.	Interior finish improper	Visual	03009
110.	Metal defective (crack, void, etc.)	Visual	03010
Minor:			
201.	Evidence of poor workmanship (See 3.8)	Visual	03011

4.3.2.4 Boattail (prior to protective coating)(See Dwg. D-8875891 covering a detail of Dwg. F-8875940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Radius behind external thread	Gage	04001

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102.	Pitch diameter of external thread	Gage	04002
103.	Major diameter of external thread, min	Gage	04003
104.	Perpendicularity of shoulder behind external thread with pitch diameter of thread	Gage	04004
105.	Length to shoulder behind external thread from forward face	Gage	04005
106.	Width of relief behind external thread	Gage	04006
107.	Length from start of external thread to shoulder, min	Gage	04007
108.	Scoring of band seat missing	Visual	04008
109.	Interior finish improper (cut, tool marks, burr, etc.)	Visual	04009
110.	Metal defective (crack, void, etc.)	Visual	04010
111.	Tool mark, crack or undercut on external thread relief, radius behind thread relief or shoulder above thread relief	Visual	04011
112.	External thread damaged (gouged, nicked, etc.)	Visual	04012
Minor:			
201.	Diameter of relief behind external thread, min	Gage	04013
202.	Diameter of external thread pilot, min	Gage	04014
203.	Chamfer missing or improper (when applicable, as per Note 9)	Visual	04015
204.	Evidence of poor workmanship (See 3.8)	Visual	04016

4.3.2.5 Forward body and sleeve assembly (See Dwg. F-8875937 covering a detail of Dwg. F-8875940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:			
101.	Perpendicularity of rear face with pitch diameter of thread	Gage	05001
102.	Depth to sleeve	Gage	05002
103.	Effective length of rear thread, min.	Gage	05003
104.	Interior finish improper	Visual	05004
Minor:			
201.	Protective finish with bare spot	Visual	05005
202.	Evidence of poor workmanship on interior walls (See 3.8)	Visual	05006
203.	Threads damaged	Visual	05007

4.3.2.6 Body assembly (prior to banding) (See Dwg. F-8875940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	100% Inspection		
101.	Forward body fails to contact boattail around circumference (gap)	Visual	06001
102.	Angle or radius to form dovetail in band seat sidewall missing (forward and rear sidewalls)	Visual	06002
Major:	(Inspection as per 4.3.2)		
103.	Boattail turns when min. torque is applied in tightening direction (See 4.3.3.12)	Test	06003

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104.	Width of band seat	Gage	06004
105.	Diameter of band seat	Gage	06005
106.	Concentricity of band seat diameter with small outside diameter	Gage	06006
107.	Bourrelet fails to overlap body around entire circumference	Gage	06007
108.	Depth of pin hole, max (6 holes)	Gage	06008
109.	Diameter of pin hole, max (6 holes)	Gage	06009
110.	Distance between pin hole and base end (6 holes)	Gage	06010
111.	Ribs or rib scores in band seat missing or improper (scores do not extend to bottom of rib)	Visual	06011
Minor:			
201.	Evidence of poor workmanship in band seat (See 3.8)	Visual	06012
202.	Pin hole improperly spaced (6 holes)	Visual	06013

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4.3.2.7 Body assembly (prior to painting) (See Dwg. F-8875940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Concentricity of inside diameter with small outside diameter (points of measurement to include forward plate seat area in bottom)	Gage	07001
102.	Wall thickness of nose in indicated area, min.	Gage	07002
103.	Depth of cavity	Gage	07003
104.	Pitch diameter of nose thread	Gage	07004
105.	Minor diameter of nose thread	Gage	07005
106.	Pitch diameter of rear thread	Gage	07006
107.	Minor diameter of rear thread	Gage	07007
108.	Diameter at nose end	Gage	07008
109.	Length of full thread, min. (nose end)	Gage	07009
110.	Distance from nose end to end of bourrelet	Gage	07010
111.	Total length	Gage	07011
112.	Distance from rear end to datum of boattail	Gage	07012
113.	Length of full thread, min (rear end)	Gage	07013
114.	Inside diameter of cavity from base end of boattail to shoulder in bottom area (Note 22)	Gage	07014

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115.	Concentricity of ogive near nose end with small outside diameter	Gage	07015
116.	Concentricity of pitch diameter of nose thread with small outside diameter including perpendicularity with nose face	Gage	07016
117.	Concentricity of large outside diameter with small outside diameter	Gage	07017
118.	Concentricity of datum diameter on ogive with small outside diameter	Gage	07018
119.	Concentricity of indicated area on boattail taper with small outside diameter	Gage	07019
120.	Concentricity of pitch diameter of boattail thread with small outside diameter including perpendicularity of rear end face	Gage	07020
121.	Concentricity of second largest rotating band diameter with small outside diameter	Gage	07021
122.	Concentricity of largest rotating band diameter with small outside diameter	Gage	07022
123.	Largest diameter of rotating band	Gage	07023
124.	Second largest diameter of rotating band	Gage	07024
125.	Third largest diameter of rotating band	Gage	07025
126.	Width of rotating band	Gage	07026

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127.	Distance from base of boattail to front end of rotating band	Gage	07027
128.	Diameter of rotating band relief	Gage	07028
129.	Diameter of rotating band groove	Gage	07029
130.	Large outside diameter	Gage	07030
131.	Small outside diameter	Gage	07031
132.	Wall variation, max.	Gage	07032
133.	Distance from nose end to datum diameter on ogive	Gage	07033
134.	Step in interior wall at body-sleeve joint near bottom of cavity (or at sleeve-boattail joint)(could impede movement of forward plate or cargo)	Visual-Manual	07034
135.	Groove pin missing or above flush (6 pins)	Visual-Manual	07035
136.	Groove pin broken through inside wall of boattail	Visual-Manual	07036
137.	Silicone compound missing from nose end and base end threads	Visual	07037

Minor:

201.	Evidence of poor workmanship (See 3.8)	Visual	07038
202.	Discoloration due to corrosive action	Visual	07039

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4.3.2.8 Body assembly (after painting and less lifting plug) (See Dwg. F-8875940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Bourrelet diameter, max.	Gage	08001
102.	Total length, max.	Gage	08002
103.	Letter "A" (indicating air test acceptance) missing	Visual	08003
104.	Rotating band missing or damaged	Visual	08004
105.	Base plug or its gasket missing	Visual	08005
Minor:			
201.	Paint not dry	Tactile	08006
202.	Protective coating with bare spot (except slight scratches)	Visual	08007
203.	Marking misleading or unidentifiable	Visual	08008
204.	Evidence of poor workmanship (See 3.8)	Visual	08009

4.3.2.9 Base plug (prior to painting) (See Dwg. D-8875892).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Pitch diameter of thread	Gage	09001
102.	Major diameter of thread, min.	Gage	09002
103.	Perpendicularity of shoulder face with pitch diameter of thread	Gage	09003

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104.	Concentricity of outside diameter with pitch diameter of thread	Gage	09004
105.	Outside diameter	Gage	09005
106.	Length of thread	Gage	09006
107.	Diameter of groove	Gage	09007
108.	Length to shoulder	Gage	09008

Minor:

201.	Diameter of wrench holes	Gage	09009
202.	Depth of wrench holes	Gage	09010
203.	Total length	Gage	09011
204.	Chamfer missing	Visual	09012
205.	Angle missing	Visual	09013
206.	Evidence of poor workmanship (See 3.8)	Visual	09014

4.3.2.10 Plug, base (See Dwg. D-8875892).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:			
201.	Base metal exposed	Visual	10001
202.	Evidence of poor workmanship (See 3.8)	Visual	10002
203.	Preservative or compound missing from threads or unpainted areas	Visual	10003

4.3.2.11 Pallet (loaded)(See Dwg. 7548346).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Strap width incorrect	Visual	11001
102.	Strapping improperly assembled	Visual	11002
103.	Connector missing or improperly assembled	Visual	11003
104.	Body assembly missing	Visual	11004
105.	Board broken or split	Visual	11005
106.	Markings for FSN or descriptive nomenclature missing, incorrect or unidentifiable	Visual	11006

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104.	Width of band seat	Gage	06004
105.	Diameter of band seat	Gage	06005
106.	Concentricity of band seat diameter with small outside diameter	Gage	06006
107.	Bourrelet fails to overlap body around entire circumference	Gage	06007
108.	Depth of pin hole, max (6 holes)	Gage	06008
109.	Diameter of pin hole, max (6 holes)	Gage	06009
110.	Distance between pin hole and base end (6 holes)	Gage	06010
111.	Ribs or rib scores in band seat missing or improper (scores do not extend to bottom of rib)	Visual	06011
Minor:			
201.	Evidence of poor workmanship in band seat (See 3.8)	Visual	06012
202.	Pin hole improperly spaced (6 holes)	Visual	06013

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4.3.2.7 Body assembly (prior to painting) (See Dwg. F-8875940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Concentricity of inside diameter with small outside diameter (points of measurement to include forward plate seat area in bottom)	Gage	07001
102.	Wall thickness of nose in indicated area, min.	Gage	07002
103.	Depth of cavity	Gage	07003
104.	Pitch diameter of nose thread	Gage	07004
105.	Minor diameter of nose thread	Gage	07005
106.	Pitch diameter of rear thread	Gage	07006
107.	Minor diameter of rear thread	Gage	07007
108.	Diameter at nose end	Gage	07008
109.	Length of full thread, min. (nose end)	Gage	07009
110.	Distance from nose end to end of bourrelet	Gage	07010
111.	Total length	Gage	07011
112.	Distance from rear end to datum of boattail	Gage	07012
113.	Length of full thread, min (rear end)	Gage	07013
114.	Inside diameter of cavity from base end of boattail to shoulder in bottom area (Note 22)	Gage	07014

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115.	Concentricity of ogive near nose end with small outside diameter	Gage	07015
116.	Concentricity of pitch diameter of nose thread with small outside diameter including perpendicularity with nose face	Gage	07016
117.	Concentricity of large outside diameter with small outside diameter	Gage	07017
118.	Concentricity of datum diameter on ogive with small outside diameter	Gage	07018
119.	Concentricity of indicated area on boattail taper with small outside diameter	Gage	07019
120.	Concentricity of pitch diameter of boattail thread with small outside diameter including perpendicularity of rear end face	Gage	07020
121.	Concentricity of second largest rotating band diameter with small outside diameter	Gage	07021
122.	Concentricity of largest rotating band diameter with small outside diameter	Gage	07022
123.	Largest diameter of rotating band	Gage	07023
124.	Second largest diameter of rotating band	Gage	07024
125.	Third largest diameter of rotating band	Gage	07025
126.	Width of rotating band	Gage	07026

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127.	Distance from base of boattail to front end of rotating band	Gage	07027
128.	Diameter of rotating band relief	Gage	07028
129.	Diameter of rotating band groove	Gage	07029
130.	Large outside diameter	Gage	07030
131.	Small outside diameter	Gage	07031
132.	Wall variation, max.	Gage	07032
133.	Distance from nose end to datum diameter on ogive	Gage	07033
134.	Step in interior wall at body-sleeve joint near bottom of cavity (or at sleeve-boattail joint)(could impede movement of forward plate or cargo)	Visual-Manual	07034
135.	Groove pin missing or above flush (6 pins)	Visual-Manual	07035
136.	Groove pin broken through inside wall of boattail	Visual-Manual	07036
137.	Silicone compound missing from nose end and base end threads	Visual	07037

Minor:

201.	Evidence of poor workmanship (See 3.8)	Visual	07038
202.	Discoloration due to corrosive action	Visual	07039

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4.3.3.4 Weight of body assembly (See Dwg. C-8875940) (Code No. 13001).-This inspection shall be performed one hundred percent (100%). Any body assembly which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5 Air pressure test of body assembly (See Dwg. C-8875940) Major defect (Code No. 14001).-This inspection shall be performed one-hundred percent. Any body assembly which fails to comply with the requirements specified on the applicable drawing shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.6 Rotating band clearance and sidewall contact (See 3.3 and Dwg. C-8875940) Major defect, Code No. 15001.-One (1) body assembly shall be selected from each eight (8) hours' production, or fraction thereof, from each banding machine in operation and subjected to this test. If any body assembly fails to comply with the requirements specified on the applicable drawing, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.7 Tension test of rotating band (See 3.4, Dwg. C-8875940) Major defect, Code No. 16001.-One (1) body assembly shall be selected from each eight (8) hours' production, or fraction thereof, from each banding machine in operation and the band subjected to this test. If any band fails to comply with the mechanical properties requirements (tensile strength and elongation) specified on the applicable drawing, it shall be classed defective and the quantity of bodies banded on that machine since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance may be used for this test.

4.3.3.7.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from that quantity represented by each failing band and tested as specified in 4.4.4. If any rotating band fails to comply with the applicable requirements, that quantity of body assemblies represented by the original failing sample shall be rejected.

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4.3.3.8 Bend test of rotating band (See 3.5, Dwg. F-8875940 and MIL-S-20426) Major defect, Code No. 17001.-One (1) body assembly shall be selected from each eight (8) hours' production, or fraction thereof, from each banding machine in operation and the band subjected to this test. If any band fails to comply with the bend test requirements specified, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance or tension test may be used for this test.

4.3.3.8.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from that quantity represented by each failing band and tested as specified in 4.4.5. If any rotating band fails to comply with the applicable requirements, the quantity of body assemblies represented by the original failing sample shall be rejected.

4.3.3.9 Tension test of sleeve (See Dwg. C-8875890), Major defect (Code No. 18001).-One (1) sleeve shall be selected from each lot of five hundred (500) sleeves, or portion thereof, and an additional sleeve selected for each additional quantity of five hundred (500) sleeves. If any sleeve fails to comply with the mechanical properties requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.3.10 Tension tests of boattail and forward body (prior to machining)(See Dwgs. D-8875891 and F-8875889, respectively) Major defect (Code No. 19001 boattail; 20001 forward body).-Boattails and forward bodies shall be selected from each mill heat of steel (See 4.3.1.1) in accordance with Table I and coupons cut therefrom. If any test coupon fails to comply with the mechanical properties requirements specified on the applicable drawing, that quantity of boattails or bodies represented by the sample coupon shall be rejected. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

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TABLE I

No. of Boattails or Bodies in Mill Heat Lot	Sample Size
50-200	2 of each component
201-300	3
301-400	4
401-500	5
501-600	6
601-700	7
701-800	8
801-900	9
Over 900	10

If more than one heat treatment or stress-relief batch is included in the mill-heat of steel, or when the heat of steel is processed in two (2) or more batch-type furnaces, the sample components shall be selected in such a manner as to be representative of the greatest number of batches possible within the mill heat. When heat treatment or stress relief is performed in a continuously operated, conveyor-type furnace, the samples shall be selected in a sequence that will be representative, to the maximum extent, of the entire mill-heat lot in process.

4.3.3.11 Tension test of base plug (See Dwg. D-8875892) Major defect (Code No. 21001).-Sample coupons of the base plug shall be taken from each mill-heat of steel in accordance with the requirements specified in 4.3.3.10 and Table I and subjected to the tension test. If any test coupon fails to comply with the mechanical properties requirements specified on the applicable drawing, that quantity of base plugs represented by the failing coupon shall be rejected. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.12 Minimum torque of boattail (See Dwg. F-8875940) Major defect, Code No. 06003.-In place of the normal sampling procedure associated with the Classification of defects, and with the approval of the Contracting Officer, the following inspection procedure may be used: at the start of each day, and whenever a new piece of equipment is put into operation, five (5) body assemblies shall be selected from each piece of equipment for determination of minimum assembly torque of the boattail for qualifying the equipment. In addition, five (5) assemblies per machine shall be selected from each hour's production, or fraction thereof, for the determination of the

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minimum boattail assembly torque as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assemblies shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment having produced the defective assembly shall be removed from production. Further, that portion of production since the last equipment check shall be returned to the contractor for corrective action. All equipment removed from production because of some fault, may, after corrective action, be returned to production providing it passes the qualification test above.

4.3.3.13 Salt spray test of base plug (See Dwg. D-8875892) Major defect (Code No. 22001).-Ten (10) base plugs shall be selected from each lot for this test. The test shall be performed as specified in 4.4.10. If any base plug fails to comply with the specified requirements, the lot shallbe rejected and the corrective action specified in TT-C-490 shall be instituted.

4.3.3.14 Proving ground ballistic tests (See 3.6 and 6.8) Major defect, Code No. 23001.

4.3.3.14.1 Initial production.-Following acceptance of the first article sample, specified in 4.2.2, the following shall apply:

a. From the initial two-hundred (200) body assemblies produced, thirty-six (36) body assemblies shall be selected, forwarded to the designated loading facility, and ballistically tested in the following zones:

Zone 1 - Ten (10). (See 4.4.9.1).
Zone 7 - Ten (10). Five (5) of the rounds shall be subjected to the five-foot drop test prior to firing (See 4.4.9.2).
Zone 7X - Fifteen (15) (See 4.4.9.3). (one (1) round shall be for replacement. If this round is not used, it shall be fired at Zone 7X).
The test shall be performed as specified in 4.4.9.
If the sample fails to comply with the specified requirements, the lot shall be rejected, production of the body assemblies shall be suspended and corrections made, as specified by the Contracting Officer.

b. From the next five-hundred (500) body assemblies produced, thirty-six (36) body assemblies shall be selected, forwarded to the designated loading facility, and ballistically tested in accordance with

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107.	Grommet missing from body assembly	Visual	11007
108.	Lifting plug missing	Visual	11008

Minor:

201.	Marking, other than above, missing, incorrect or unidentifiable	Visual	11009
202.	Evidence of poor workmanship (See 3.8)	Visual	11010

4.3.3 Testing.

4.3.3.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.2 Tool control inspection covering the following characteristics:

a. Forward body (Dwg. F-8875889)
   (1) Chamfer at rear end, min.
   (2) Radius at bottom of large (sleeve) cavity.
   (3) Radius at bottom of smaller cavity (forward plate seat).

b. Body assembly (Dwg. F-8875940)
   (1) Radius at bottom of smaller cavity (forward plate seat)(when machined at assembly)

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(2) Flat at top edge of vertical sidewall of rotating band groove, max. (2 places-on boattail and rear face of forward body).

(3) Angle of rotating band groove sidewall (2 places - on boattail and rear face of forward body)

At the start of production, after every four (4) hours of use, and whenever a change or modification of punching tools is made during production, the required characteristic measurements shall be obtained from that tool. Determination of characteristics through the use of wax impressions is permissible. The contractor's tolerance limits for control of these characteristics shall be approved by the procuring activity. When a measurement falls outside the contractor's approved tolerance limits, twenty-five (25) forward bodies, or body assemblies (as applicable) from the latter portion of production from the tool since the last acceptable measurement was taken, shall be measured for the required characteristics. The use of wax impressions is a permissible means of obtaining these characteristics for measurement. If any characteristic measurement taken in this manner falls outside the component drawing tolerance, the forward bodies and body assemblies produced from that tool since the last acceptable measurement was obtained shall be inspected one hundred percent for that characteristic by the contractor, to remove any forward bodies and body assemblies which are outside drawing tolerance.

In addition, the required characteristic measurements shall be taken from five (5) forward bodies and five (5) body assemblies from each eight (8) hour's production, or portion thereof. The use of wax impressions is a permissible means of obtaining these measurements. If any characteristic measurement falls outside of the drawing requirement, the forward bodies and body assemblies produced since the last satisfactory measurement was obtained shall be inspected one hundred percent by the contractor to remove any forward bodies and body assemblies which do not comply with the applicable drawing requirement.

4.3.3.3 Steel blooms or billets with hot-sheared ends (See 3.7 and Dwgs. D-8875892, D-8875891 and F-8875889), Major defect, Code No. 12001 (when applicable).-A sample of fifty (50) steel blooms or billets with hot-sheared ends shall be taken from that quantity which are to be used in the manufacture of body assemblies for the first article quantity and examined visually prior to cutting the slugs or blanks for the die operation. If one (1) or more billets fail to exhibit hot-sheared ends, the production tooling shall be corrected until it is demonstrated that the tooling is capable of using billets with hot sheared ends.

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4.3.3.4 Weight of body assembly (See Dwg. C-8875940) (Code No. 13001).-This inspection shall be performed one hundred percent (100%). Any body assembly which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5 Air pressure test of body assembly (See Dwg. C-8875940) Major defect (Code No. 14001).-This inspection shall be performed one-hundred percent. Any body assembly which fails to comply with the requirements specified on the applicable drawing shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.6 Rotating band clearance and sidewall contact (See 3.3 and Dwg. C-8875940) Major defect, Code No. 15001.-One (1) body assembly shall be selected from each eight (8) hours' production, or fraction thereof, from each banding machine in operation and subjected to this test. If any body assembly fails to comply with the requirements specified on the applicable drawing, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.7 Tension test of rotating band (See 3.4, Dwg. C-8875940) Major defect, Code No. 16001.-One (1) body assembly shall be selected from each eight (8) hours' production, or fraction thereof, from each banding machine in operation and the band subjected to this test. If any band fails to comply with the mechanical properties requirements (tensile strength and elongation) specified on the applicable drawing, it shall be classed defective and the quantity of bodies banded on that machine since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance may be used for this test.

4.3.3.7.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from that quantity represented by each failing band and tested as specified in 4.4.4. If any rotating band fails to comply with the applicable requirements, that quantity of body assemblies represented by the original failing sample shall be rejected.

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4.3.3.8 Bend test of rotating band (See 3.5, Dwg. F-8875940 and MIL-S-20426) Major defect, Code No. 17001.-One (1) body assembly shall be selected from each eight (8) hours' production, or fraction thereof, from each banding machine in operation and the band subjected to this test. If any band fails to comply with the bend test requirements specified, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance or tension test may be used for this test.

4.3.3.8.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from that quantity represented by each failing band and tested as specified in 4.4.5. If any rotating band fails to comply with the applicable requirements, the quantity of body assemblies represented by the original failing sample shall be rejected.

4.3.3.9 Tension test of sleeve (See Dwg. C-8875890), Major defect (Code No. 18001).-One (1) sleeve shall be selected from each lot of five hundred (500) sleeves, or portion thereof, and an additional sleeve selected for each additional quantity of five hundred (500) sleeves. If any sleeve fails to comply with the mechanical properties requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.3.10 Tension tests of boattail and forward body (prior to machining)(See Dwgs. D-8875891 and F-8875889, respectively) Major defect (Code No. 19001 boattail; 20001 forward body).-Boattails and forward bodies shall be selected from each mill heat of steel (See 4.3.1.1) in accordance with Table I and coupons cut therefrom. If any test coupon fails to comply with the mechanical properties requirements specified on the applicable drawing, that quantity of boattails or bodies represented by the sample coupon shall be rejected. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

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TABLE I

No. of Boattails or Bodies in Mill Heat Lot	Sample Size
50-200	2 of each component
201-300	3
301-400	4
401-500	5
501-600	6
601-700	7
701-800	8
801-900	9
Over 900	10

If more than one heat treatment or stress-relief batch is included in the mill-heat of steel, or when the heat of steel is processed in two (2) or more batch-type furnaces, the sample components shall be selected in such a manner as to be representative of the greatest number of batches possible within the mill heat. When heat treatment or stress relief is performed in a continuously operated, conveyor-type furnace, the samples shall be selected in a sequence that will be representative, to the maximum extent, of the entire mill-heat lot in process.

4.3.3.11 Tension test of base plug (See Dwg. D-8875892) Major defect (Code No. 21001).-Sample coupons of the base plug shall be taken from each mill-heat of steel in accordance with the requirements specified in 4.3.3.10 and Table I and subjected to the tension test. If any test coupon fails to comply with the mechanical properties requirements specified on the applicable drawing, that quantity of base plugs represented by the failing coupon shall be rejected. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.12 Minimum torque of boattail (See Dwg. F-8875940) Major defect, Code No. 06003.-In place of the normal sampling procedure associated with the Classification of defects, and with the approval of the Contracting Officer, the following inspection procedure may be used: at the start of each day, and whenever a new piece of equipment is put into operation, five (5) body assemblies shall be selected from each piece of equipment for determination of minimum assembly torque of the boattail for qualifying the equipment. In addition, five (5) assemblies per machine shall be selected from each hour's production, or fraction thereof, for the determination of the

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minimum boattail assembly torque as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assemblies shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment having produced the defective assembly shall be removed from production. Further, that portion of production since the last equipment check shall be returned to the contractor for corrective action. All equipment removed from production because of some fault, may, after corrective action, be returned to production providing it passes the qualification test above.

4.3.3.13 Salt spray test of base plug (See Dwg. D-8875892) Major defect (Code No. 22001).-Ten (10) base plugs shall be selected from each lot for this test. The test shall be performed as specified in 4.4.10. If any base plug fails to comply with the specified requirements, the lot shallbe rejected and the corrective action specified in TT-C-490 shall be instituted.

4.3.3.14 Proving ground ballistic tests (See 3.6 and 6.8) Major defect, Code No. 23001.

4.3.3.14.1 Initial production.-Following acceptance of the first article sample, specified in 4.2.2, the following shall apply:

a. From the initial two-hundred (200) body assemblies produced, thirty-six (36) body assemblies shall be selected, forwarded to the designated loading facility, and ballistically tested in the following zones:

Zone 1 - Ten (10). (See 4.4.9.1).
Zone 7 - Ten (10). Five (5) of the rounds shall be subjected to the five-foot drop test prior to firing (See 4.4.9.2).
Zone 7X - Fifteen (15) (See 4.4.9.3). (one (1) round shall be for replacement. If this round is not used, it shall be fired at Zone 7X).
The test shall be performed as specified in 4.4.9.
If the sample fails to comply with the specified requirements, the lot shall be rejected, production of the body assemblies shall be suspended and corrections made, as specified by the Contracting Officer.

b. From the next five-hundred (500) body assemblies produced, thirty-six (36) body assemblies shall be selected, forwarded to the designated loading facility, and ballistically tested in accordance with

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"a"(above). The test shall be performed as specified in 4.4.9. If the sample fails to comply with the requirements specified, the lot shall be rejected, production of the body assemblies shall be suspended and corrections made, as specified by the Contracting Officer.

c. From the next five-hundred (500) body assemblies produced, thirty-six (36) body assemblies shall be selected, forwarded to the designated loading facility, and tested as specified in "a" (above).

4.3.3.14.2 Regular production.-Following acceptance of all initial production lots as specified in 4.3.3.14.1 a, b, & c, the following shall apply: from each lot of body assemblies produced, twenty-four (24) body assemblies shall be selected, forwarded to the designated loading facility, and ballistically tested in the following zones:

Zone 1 - Four (4). (See 4.4.9.1).
Zone 7 - Five (5). The five rounds shall be subjected to the five-foot drop test prior to firing. (See 4.4.9.2).
Zone 7X - Fourteen (14). (See 4.4.9.3). One (1) round shall be for replacement. If it is not used, it shall be fired at Zone 7X). The test shall be performed as specified in 4.4.9. If the sample fails to comply with the requirements specified, the lot shall be rejected, production of the body assemblies shall be suspended, and corrections made, as specified by the Contracting Officer.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3 (See 6.3).

4.4 Test methods and procedures.

4.4.1 Weight of body assembly.-Each body assembly shall be weighed using methods and equipment in accordance with 4.3.4.

4.4.2 Air pressure test of body assembly.-All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated at the start of each shift and, thereafter, at intervals of not more than four (4) hours of continuous operation. It shall also

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be calibrated prior to restarting the operation after a layoff of three (3) hours or more. If it is found that the gage is out of calibration in such a direction that the test pressure was actually lower than that specified on the drawing, body assemblies tested since the last satisfactory calibration shall be retested after the gage error has been corrected. All test and calibration instructions, equipment and procedures shall be approved in accordance with 4.3.4.

The test fixture shall be constructed in such a manner that the pressure specified on the applicable drawing will be applied to the inside of the body assembly. The body assembly shall then be isolated from the pressure supply. A leaking assembly shall be evidenced by a decrease in gage pressure as the assembly remains pressurized (air supply cut-off) for the time specified on the applicable drawing. NOTE.-Use of water in performing this test, such as an immersion test, shall not be allowed.

4.4.3 Rotating band clearance (tightness).-Prior to removal, the diameter of the finished band shall be measured on three (3) diameters, 60° apart, and on two (2) points along the cylindrical portion of its length near the front and near the rear edge. This comprises six (6) diametral measurements. The points of measurement shall be marked on the bands as well as the orientation of the band on the boattail. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve (12) measurements), and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding two (2) thicknesses of the band and subtract the result from the corresponding exterior diameter of the band previously determined. The result will indicate the total diametral clearance between the band and its seat. If the band is well seated, a negative clearance, or interference, may be indicated on some of the measurements or even on the average of all measurements. When waved band ribs or irregular band profile interfere with ease of measurement, the manufacturer is authorized to machine the outside of the band in order to obtain the adequate cylindrical surface. The diameter of the machined band must exceed the bourrelet diameter of the forward body.

Sidewall contact.-Examination shall be made to determine that the band blank makes contact with the sidewalls for the minimum of 75 percent of the band seat sidewall depth.

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4.4.4 Tension test of rotating band.-The band shall be carefully removed from the body assembly and carefully flattened without hammering. The test specimen shall be cut along the direction of the flattened length and prepared in such a manner as to hold to the minimum any additional cold working. Tension test specimens shall be machined to the form and dimensions specified in MIL-S-20426. The specimens shall be tested as specified in MIL-S-20426. Observation shall be made for tensile strength and percentage of elongation.

4.4.5 Bend test of rotating band.-The band, or section thereof, of full width and thickness, except as reduced by authorized surface cut, shall be bent cold by pressure, without hammering, in the direction of curvature of the band, until it tightly encloses 180 degrees of a rod equal in diameter to the thickness of the band test specimen (within plus or minus 0.010 inch). The band shall be observed for cracks resulting from the test.

4.4.6 Tension test of sleeve.-The sleeve shall be subjected to the tension test in accordance with ASTM E 8-69 (Part 31) Method 211.1 to determine its mechanical properties (yield strength, tensile strength, and percentage of elongation).

4.4.7 Tension tests of boattail and forward body (Prior to machining).-Prior to any machining operations on the boattail and forward body, and after completion of all heat treatment operations including stress-relief, two (2) specimens (coupons) conforming to the size prescribed by ASTM E 8-69, Figure 6, Sheet-Type, shall be taken longitudinally from each side of the boattail and forward body. The specimens from the boattail shall be taken from the area of the band seat 180 degrees apart. The gage length of these specimens shall be from within the width of the band seat area. The specimens from the forward body shall be taken longitudinally from the bourrelet area 180 degrees apart. The test shall be conducted in accordance with ASTM E 8-69. (Part 31). Yield strength shall be determined by the offset method. The offset shall be 0.1 percent (0.001 inch per inch of gage length). In addition, the yield point shall be determined by the divider method. Identity of each heat-treated lot shall be maintained until minimum mechanical properties (yield strength and percentage of elongation) have been verified by the Government inspector.

4.4.8 Tension test of base plug.-Specimens shall be cut from base plugs in such a manner that the longitudinal axis of the specimen will be parallel to direction of grain. Mechanical properties (yield strength, tensile strength, and percentage of

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elongation) shall be determined in accordance with ASTM E 8-69. Part 31). Yield strength may be determined by either the offset method or extension-under-load method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length). In addition, the yield point shall be determined by the divider method. Identity of each heat treated lot shall be maintained until all minimum mechanical properties have been verified by the Government inspector.

4.4.9 Proving ground ballistic tests.-Ballistic tests specified in 4.2.2.1 and 4.3.3.14 shall be conducted at a Government-owned proving ground. A suitable recovery area shall be prepared prior to firing any rounds. The area surface shall be such as to minimize the effects of impact on the body assembly to enable recovery of the body assembly intact and its hardware for subsequent examination and evaluation. The fuze setting on each round shall be photographed prior to firing (for identification purposes, body sample number shall appear just below the fuze.) The fuze, with Expulsion Charge Assembly, shall be set to achieve a minimum burst height of fifteen hundred (1500) feet. Photographic equipment shall be positioned near the gun muzzle to record the round as it emerges from the gun. All rounds shall be fired from a weapon for which they are standard. Velocity and chamber pressure shall be recorded for each round fired. Examinations shall be made of recovered body assemblies, support rods, forward plates and base plugs and all photographs to determine any evidence of metal parts failure and to determine compliance with the requirements of 3.6.

All test results and other pertinent information shall be recorded (See 6.5) and shall include the following:

a. Evidence of failure to comply with the requirements detailed in 3.6.
b. Rupturing or cracking of ogive in wall area at bottom of large cavity in forward body.
c. Probable cause of defect. (ballistically caused, ground impact, manufacturing defect, etc.).
d. Chamber pressure for each round.
e. Velocity for each round.
f. Analysis of cargo functioning. (H.E.-loaded rounds only).
g. Evidence of deformed or damaged support rods.
h. Unusual sounds at gun when firing.
i. Evidence of cargo movement. (rotation marks on sidewall of body assembly).
j. Uniformity of engraving on outside diameter of forward body near ogive. (when present).
k. Any deviation from flight pattern or trajectory.
l. Impact pattern.

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4.4.9.1 Zone 1.-The ten (10) rounds (or 4 rounds when testing regular production) shall be fired at Zone 1 using the appropriate Zone 1 propelling charge increments. Firings shall be in accordance with 4.4.9.

4.4.9.2 Zone 7.-The ten (10) rounds (or 5 rounds when testing regular production) shall be fired at Zone 7 using the appropriate Zone 7 propelling charge increments. Firings shall be in accordance with 4.4.9. Prior to firing, five (5) of the rounds, inert-loaded, and with grommet and lifting plug assembled, (all 5 rounds when testing regular production) shall be subjected to a five-foot drop test. The projectile shall be dropped five feet (lowest point of the projectile to point of impact) onto a steel plate which is solidly supported by concrete. The orientation of the projectile shall be at an axis 45 degrees from the vertical, with the base down. The projectile will be attached by means of a chain or cable at the lifting plug to prevent the projectile from tumbling after impact. The armor plate upon which impact occurs shall have a minimum thickness of 3 inches, a Brinnel hardness of at least 207, and shall be solidly supported in a horizontal plane over its entire bearing area by a minimum thickness of 24 inches of concrete. The steel plate shall have a smooth and flat surface. Upon completion of the drop test, the projectile shall be examined for evidence of separation, loosening, or fracture of any body assembly or base plug.

4.4.9.3 Zone 7X.-The fifteen (15) rounds (or 14 rounds when on regular production), H.E.-loaded, shall be fired at Zone 7X. Firings shall be in accordance with 4.4.9. Prior to firing, the rounds shall be temperature-conditioned for not less than 24 hours at the temperature specified ± 5°:

Seven (7) of the rounds shall be conditioned to -65°F.
Eight (8) of the rounds shall be conditioned to +165°F. (7 rounds when on regular production).

The rounds shall be fired using a propelling charge adjusted to give 112 percent of the rated pressure of the propelling charge. (The rated pressure appears on the data card of each propelling charge lot.)

4.4.10 Salt Spray test of Base Plug.-This test shall be performed on the base plugs as specified in ASTM B 117 (Part 7). The salt spray exposure time and the percentage of sodium chloride solution shall be as indicated on the applicable drawing. Examination shall be made for any evidence of corrosion on the painted areas. Base plugs so tested shall not be returned to any lot. (Destructive test).

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5. PREPARATION FOR DELIVERY

5.1 Packing (Level C).-The body assemblies, with base plug, shall be palleted as specified in Dwg. F-7548346.

5.2. Marking.-Marking shall be in accordance with Dwg. F-7548346.

5.3 Shipping.-When shipments from more than one lot of body assemblies, base plugs, or base plug gaskets are made at one time, such as in a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167 (See 6.9).
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

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c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (See 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion data for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to the address specified in 6.6.

6.5 Results of proving ground ballistic tests.-A copy of the ballistic test results and any other requested information relating to these tests shall be forwarded to the address specified in 6.6.

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6.6 Mechanical properties-Documentation to be submitted as specified in 4.2.4, shall be forwarded by the Contracting Officer to:

Picatinny Arsenal
Dover, New Jersey 07801
ATTN: SARPA-QA-A-S

6.7 Proving ground ballistic tests.-In order to minimize the time required by the loading facility and the proving ground, the Procuring Contracting Officer shall take all measures necessary to expedite the performance of the ballistic test phase of the first article inspection specified herein.

6.8 Special instructions to loading facility (See 6.7). The following shall apply:

a. The body assembly to be loaded for ballistics testing shall be numbered for identification purposes.

b. Prior to inserting the support rod, forward plate and base plug, these parts shall have metal-stamped on them the same number as the sample number assigned to the body assembly with which they are to be assembled.

c. The projectiles to be ballistically tested at Zones 1 and Zone 7 shall be inert-loaded. (sliders may contain live primers). The inert material may be wax, borax or other approved inert material.

d. The projectiles to be ballistically tested at Zone 7X shall be H.E.-loaded. (In order to expedite this phase of the test, inert cargo may be utilized when approved by the Contracting Officer.-See 6.7).

The loaded assemblies shall be forwarded to the Proving Ground specified by the Contracting Officer for ballistic testing as specified in 4.4.9.

6.9 Distribution of data cards.-Distribution of data cards shall include the address specified in 6.6.

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